

Self-supervised Learning in Time Series

NorwAI

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for AI Innovation



Time Series / Sensor Data

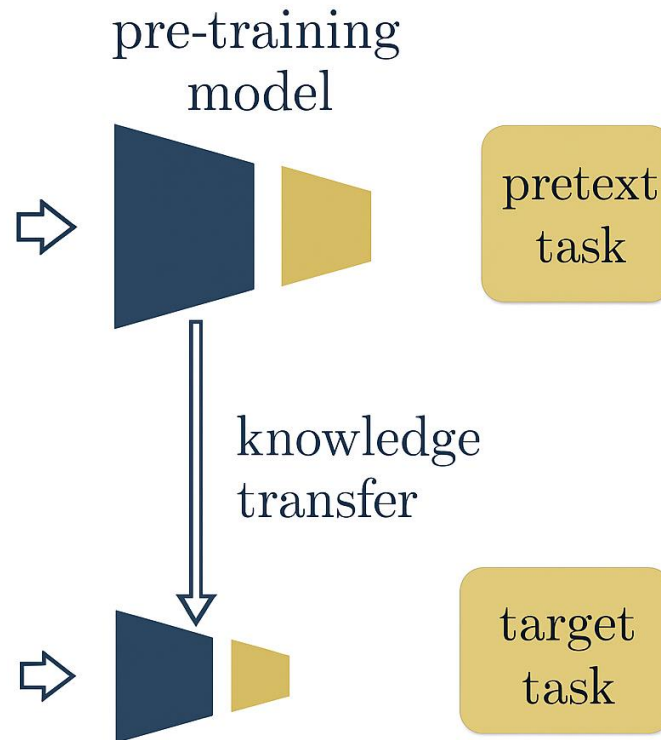
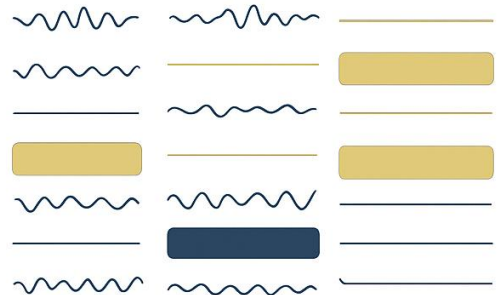


Self-Supervised Learning Motivation

dataset (no labels)



dataset (with labels)



Supervised learning

- Relies on large labeled datasets
- Expensive and domain-specific annotations
- Limited generalization

Self-Supervised learning

- Learns representations directly from data
- Scalable (unlabeled data is abundant)
- Improves transferability to downstream tasks

Self-supervised Learning in Time Series

- eMargin, LDD Workshop ECAI, 2025
- Contrast All The Time (CaTT), Main Track ECAI, 2025
- Divide and Contrast (Di-COT), Main Track ICML 2026
- Learning by Shifting (ShiFT), ECML 2026 (*Under review*)
- Context Prediction (C-Pred), NeurIPS 2026 (*Under review*)

Success criteria

- No augmentation and less domain specific assumptions
- Improved accuracy and generalization
- Substantial reduction in training time
- Compact (clusters) and transferable representations



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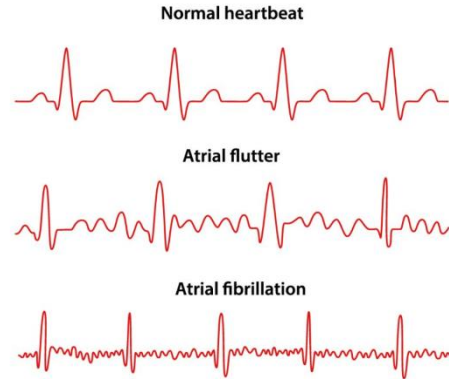
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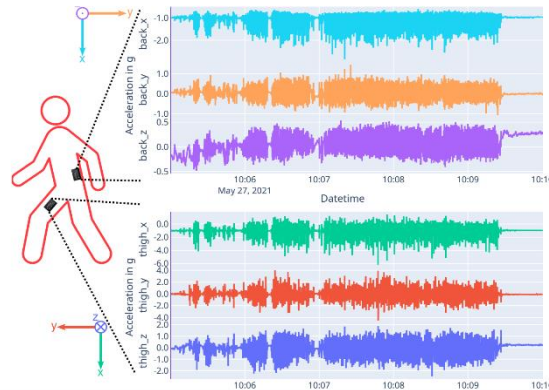
Divide and Contrast

Learning Robust Temporal Features without Augmentation

ECG

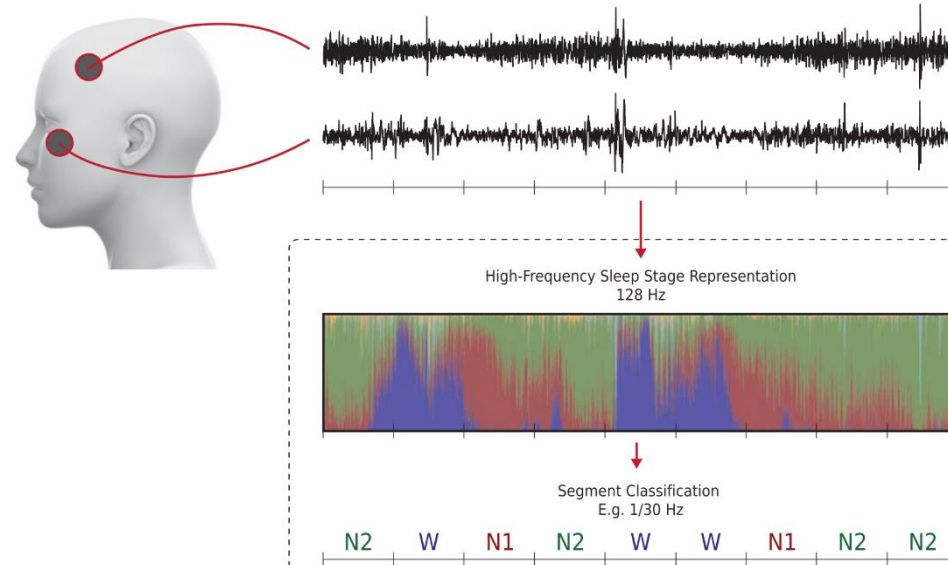


HARTH



Dataset	Train set	Test set	Subject	Sequence Length	Channel	Class
ECG	59,922	16,645	23	2,500	2	4
HARTH	23,025	2,785	22	500	6	18
PAMAP2	51,192	11,590	9	100	52	18
SKODA	22,587	5,646	1	50	64	11
SLEEP	25,612	8,910	20	3,000	1	5
WISDM2	134,614	14,421	29	40	3	6

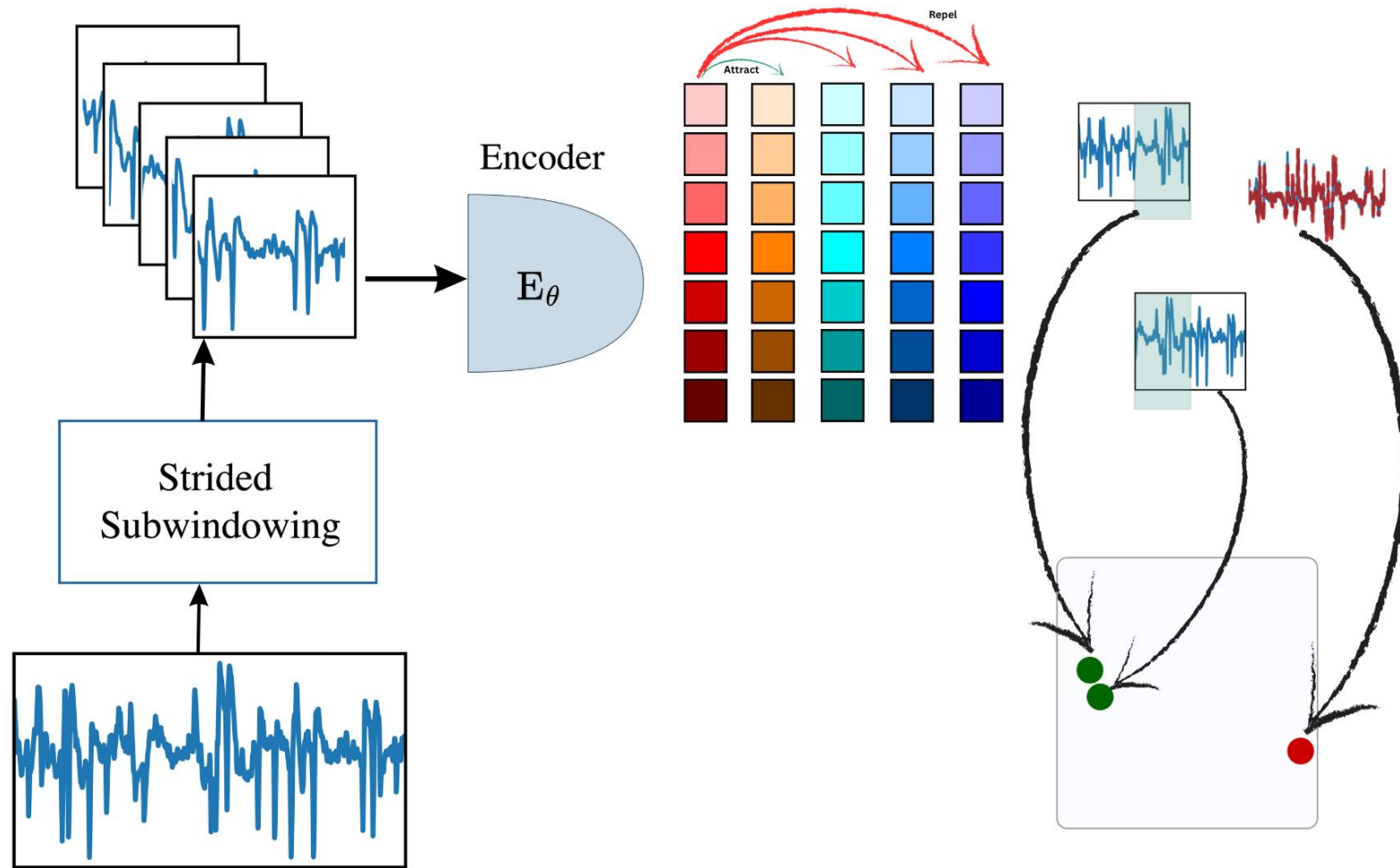
SleepEEG



source: <https://nhuopen.ntnu.no/ntnu-xmlui/handle/11250/3181556>, <https://norvinja.com/what-is-affib/>, U-sleep dataset

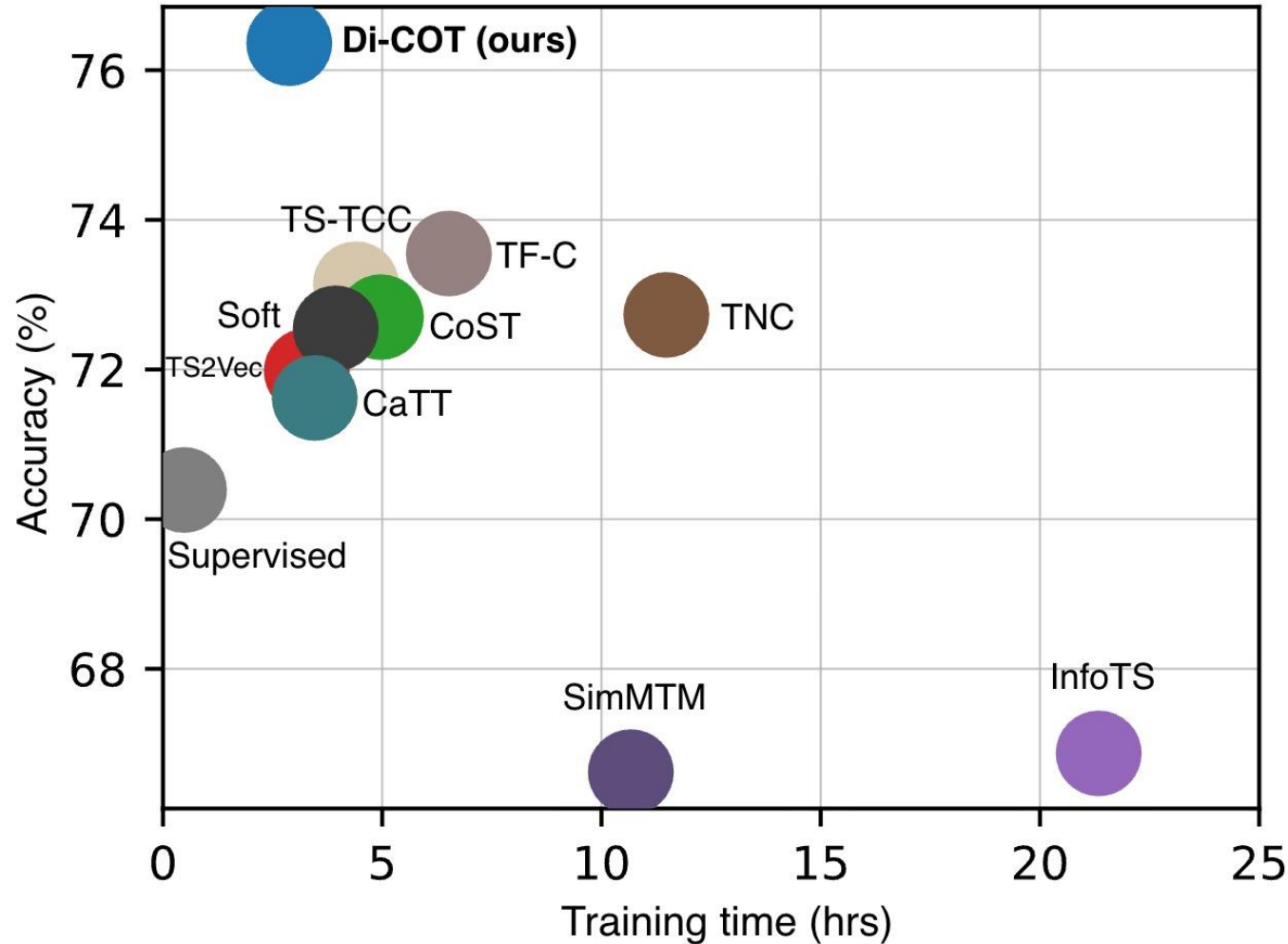
Divide and Contrast

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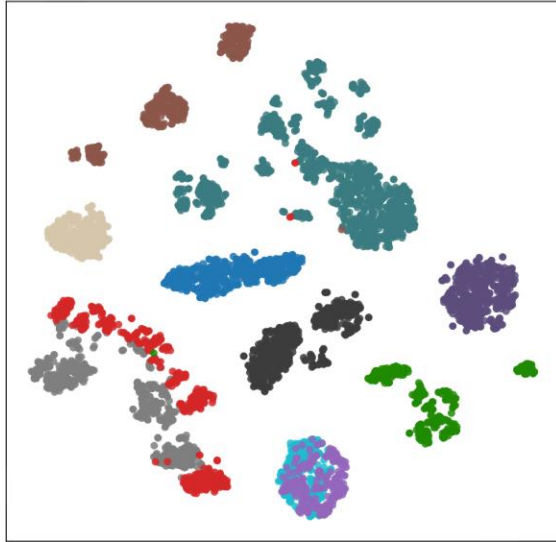


- 1% of Labeled Data
- Highest Average Accuracy
- Fastest among self-supervised method

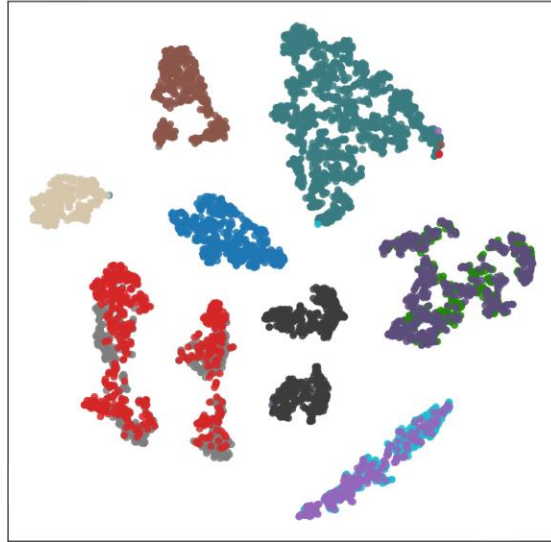
Divide and Contrast

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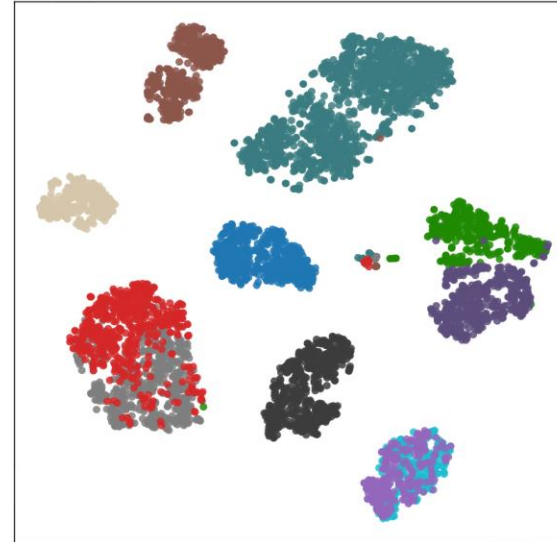
Random Init.



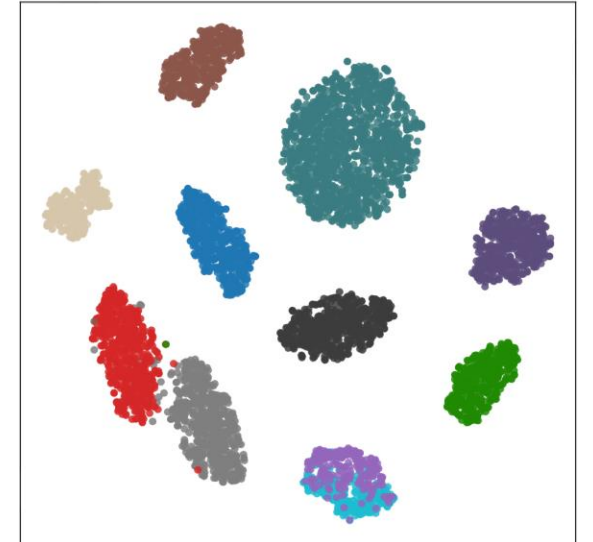
SimMTM



Supervised (*backbone*)



Di-COT (ours)

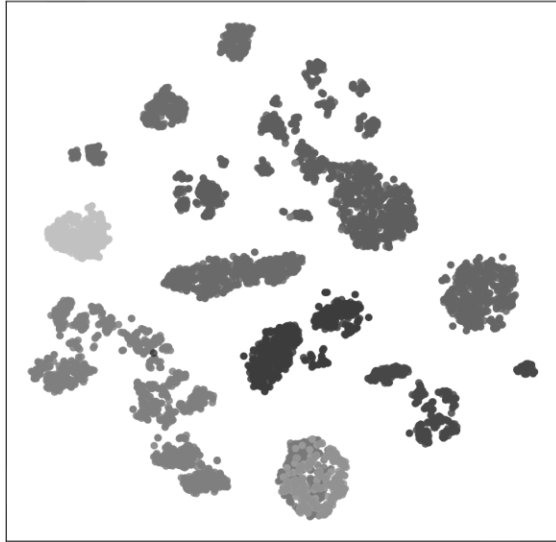


- Cleanly separates the challenging **red** and **gray** classes
- Supervised Backbone with access to full labeled train data (Limited generalization!)
- Translational invariance is *almost* all you need!

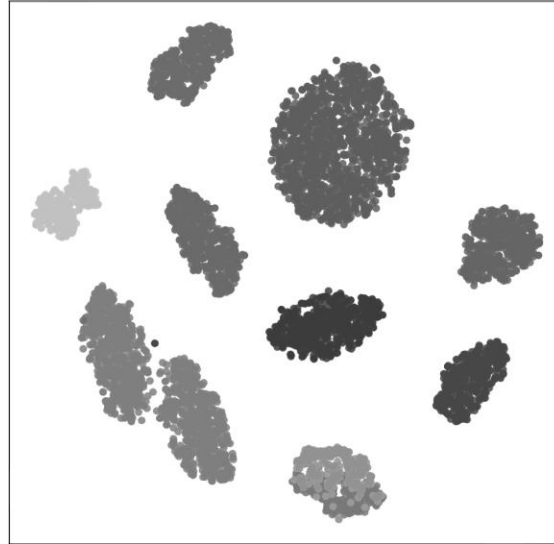
Divide and Contrast

Learning Robust Temporal Features without Augmentation

Random Init.



Di-COT (ours)



- Time series foundation model (Cluster Everything!)

- Saves time and resources for annotations while improving classification performance
- Potential for anomaly detection by uncovering hidden patterns in data without human/expert intervention
- Low resource robot deployment under critically constrained environment

Read more!



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